Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2023/2024

COURSE NAME	:	DISCRETE STRUCTURE
COURSE CODE	:	BIC 10103
PROGRAMME CODE	:	BIS/BIP/BIW/BIM
EXAMINATION DATE	:	JANUARY / FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

TERBUKA

Q1 (a) Analyse whether $\neg(\neg p \wedge q) \rightarrow (\neg q \vee r)$ is tautology, contradiction or contingency by constructing the truth table.

(7 marks)

(b) Suppose that a function, f is defined recursively by,

$$f(0) = 3$$

$$f(n+1) = 2f(n) + 3$$

Find $f(1), f(2), f(3), f(4)$ and $f(5)$.

(5 marks)

(c) Show that the following proposition, $P(n)$ is true for all positive integers by using mathematical induction.

$$\sum_{i=0}^{n-1} i(i+1) = \frac{n(n-1)(n+1)}{3}$$

(8 marks)

Q2 (a) Given $A = \{3,4,5,6\}$. Define the relation R on A as $for\ every\ x, y \in A$,

$$xRy \iff 2\ divides\ (x-y)$$

Draw the directed graph of R.

(5 marks)

(b) Given $f(x) = x^2 + 2x - 3$ and $g(x) = 3x - 1$.

(i) Find the composite function of $g \circ f$.

(3 marks)

(ii) Find the composite function of $f \circ f$.

(3 marks)

(iii) Evaluate the function of $g \circ f(4)$.

(2 marks)

(iv) Evaluate the function of $f \circ f(9)$.

(2 marks)

TERBUKA

Q3 (a) Answer **Q3(a)(i)** to **Q3(a)(iii)** according to **Figure Q3.1**.

```
In the Faculty of Computer Science and Information Technology,
the enrolment for the Discrete Structure course has been
steadily growing year by year. Currently, there are 40 students
actively participating in the Discrete Structure course.
```

Figure Q3.1

(i) Set up the recurrence relation if the number is increasing by five each year.

(1 mark)

(ii) Write down the expression of a_n in terms of n for the recurrence relation answered in **Q3(a)(i)**.

(2 marks)

(iii) Find the number of students expected to be studying Discrete Structure in five years later. Show your works.

(2 marks)

(b) A sequence of numbers is given by the recurrence relation,

$$a_{n+1} = 5 - \frac{18}{4 + a_n}, \quad n \geq 1, \quad a_2 = 0$$

Find the value of a_1 and a_3.

(3 marks)

(c) Formulate a solution to the following recurrence relations.

$$a_n = a_{n-1} + 6a_{n-2} \text{ where } a_0 = 1 \text{ and } a_1 = 8$$

(7 marks)

Q4 (a) Estimate the time complexity, $T(n)$ and Big O notation, $O(n)$ of the algorithm in **Figure Q4.1**.

```
sum = 0;
for ( i = 0; i <= n; i++)
{
for   ( j = 0; j <= n; j++)
sum = sum + j;
}
```

Figure Q4.1

(7 marks)

(b) Create an algorithm with only one looping involve using $T(n) = 3n + 1$.

(8 marks)

TERBUKA

Q5 (a) Draw a graph that represents the route that connects the residential colleges. based on **Figure Q5.1**.

A shuttle bus travels through eight residential colleges (C1, C2, C3, C4, C5, C6, C7, C8) to pick up students. The location of the colleges determines the bus route. There are two routes connecting C1 and C2. From C2, the shuttle can go to C3 directly or through two routes, one route to C4 and one route to C5 and to C3. There is only one route that connects C5, C6, and C7. In order to get directly to C7 from C2, the shuttle bus needs to pass through C5 and C6. C8 can only be access by a route through C7. During semester break, a shortcut route that connects C1, C4 and C7 can be used to shorten the journey.

Figure Q5.1

(10 marks)

(b) Describe whether the graph in **Figure Q5.2** has an Euler cycle or Euler path. Based on your answer, list out the path or the cycle.

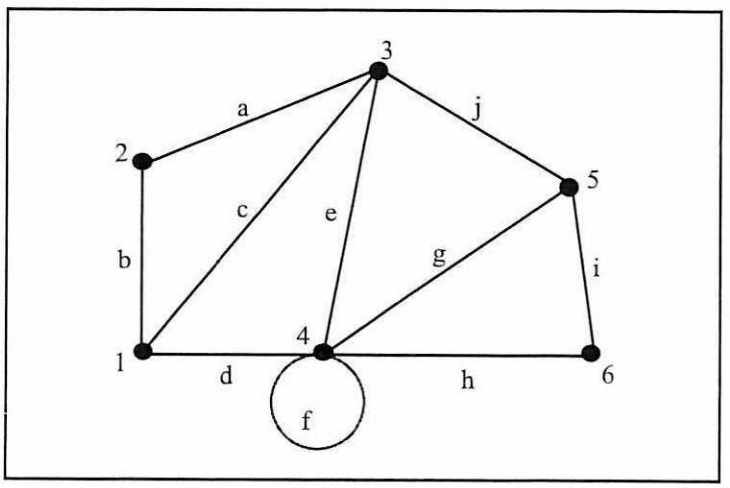


Figure Q5.2

(5 marks)

TERBUKA

- END OF QUESTIONS -